

Unit 6 – Inference for Categorical Data: Proportions  
Notes 7 – Confidence Interval for a Difference in Proportions

**Recall: Comparing Two Proportions**

Comparing two proportions based on random sampling or a randomized experiment is one of the most common situations encountered in statistical practice. We see this when:

- We want to determine whether two treatments caused the observed difference.
- We want to draw a conclusion about the population for all subjects.

**Sampling Distribution of  $\hat{p}_1 - \hat{p}_2$**

\_\_\_ = sample proportion from the first group

\_\_\_ = sample proportion from the second group

| Shape         | Center         | Spread        |
|---------------|----------------|---------------|
|               |                |               |
| As long as... | As long as.... | As long as... |

**Confidence Interval for a Difference between two Proportions**

$$\text{Statistic} \pm (\text{Critical Value}) \cdot (\text{Standard Deviation})$$

When the conditions are met, an approximate C% confidence interval for  $\hat{p}_1 - \hat{p}_2$  is

Where  $z^*$  is the critical value for the standard Normal curve with C% of its area between  $-z^*$  and  $z^*$

### Example: Two Proportion Z Interval

A recent survey wanted to compare the proportion of teenagers and adults who use social media. The first poll asked a random sample of 300 teenagers if they used social networking sites and a second random sample asked 400 adults if they used social networking sites. In these two samples, 92% of teenagers used a social networking site and 75% of adults did. Construct and interpret a 95% confidence interval for the difference between the proportions of all US teens and adults who use social-networking sites.

### On the TI-84

Stat → Tests → B: 2-Prop Z Interval

Asks for:

$x_1$  ,  $n_1$  ,  $x_2$  ,  $n_2$  , C-Level: (decimal)

Returns:

( \_\_\_\_\_ , \_\_\_\_\_ ) ,  $\hat{p}_1$  ,  $\hat{p}_2$

Test the calculator using the information from the problem we just did!

